PEER REVIEW OF THE TEMPORARY RESIDENT VISA ELIGIBILITY MODEL

Immigration, Refugees and Citizenship Canada
Immigration, Réfugiés et Citoyenneté Canada

Canada

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Executive Summary

This document provides an evaluation by the National Research Council Canada (NRC) in July 2018 of the proposed methodology for assessing the complexity of visa applications and routing them to the appropriate agents for processing. At the time of the evaluation, this methodology had not yet been implemented, and the models reviewed were not in use. This evaluation does not cover all considerations related to the Treasury Board Secretariat's Directive on Automated Decision-Making, as these were introduced in 2019.

The initiative's methodology is excellent and follows the steps necessary for a successful machine learning project. The initiative is very well adapted to different organizational risks (legal, public perception, security) while maximizing performance measures, i.e. transparency of methodology, efficiency of operations and quality of data.

The analysis provides nine recommendations and three collaborative projects with the NRC. Topics covered range from machine learning models, the working environment, algorithms as well as reproducibility of results.

1. Objectives

Immigration, Refugees and Citizenship Canada (IRCC) requires the services of NRC to conduct a scientific/methodological review of the tools and approaches that are used for the deployment of predictive models that will be use as a solution to support the Temporary Resident Volume Management (TRVM) initiative.

The rationale for conducting a scientific/methodological review by a third independent party is to ensure both the integrity, robustness and quality of the work that had been accomplished so far.

In principle, a cursory evaluation of the key phases includes the Business understanding, Data understanding, Data preparation, Modelling, Evaluation and Deployment. Since the project is relatively nascent the focus will be restricted as described in the "Scope of Work" below.

Once the cursory review is completed, both participants will determine whether a more detailed evaluation would be required.

2. Background

The Advanced Analytics Lab (AAL) is responsible for undertaking the TRVM project to support the Operations Sector in achieving client service and integrity risk management outcomes in several lines of business. The scope of this review includes China Temporary Resident Visa applicants received online.

3. Requirement

A scientific/methodological review is essentially a "peer" review process of applying expert knowledge of acceptable criteria to determine whether the tools and protocol are adequate.

A high-level report will describe what was accomplished so far including an assessment of the overall merits and identification of the tools and protocols that have been put in place.

A scientific/methodological review is a constructive process. The intent of the review is to assist IRCC to meet a minimally adequate set of criteria for the work that has been accomplished.

4. Scope of Work

The requirement is to engage the services of NRC to supply professional services expertise on the following objectives:

1. Review and assess key phases of the work undertaken so far in the following areas:
    a. Data understanding, in particular on data quality.
    b. Data preparation: data cleaning techniques and results, derived attributes and generated records, integration data approach and aggregations.
    c. Modelling: modelling technique and assumptions, test design, model build (settings, models, descriptions), assessments and revised parameter settings.
    d. Evaluation (efficiency of the model, deficiencies).

5. Tasks

The tasks required of NRC will include the following:

1. Develop an action plan outline to conduct the review
    a. Review documentation
    b. Meet key stakeholders
2. Deliverables and Acceptance Criteria:
    a. The Plan
    b. An interim Report
    c. Final Report on the detailed and comprehensive review of the method.

6. Activities

Two one-day meetings took place at IRCC on March 20th and 27th, 2018, respectively. This is the final report.

7. Analysis

The initiative's methodology is excellent and follows the steps necessary for a successful machine learning project. The initiative is very well adapted to different organizational risks (legal, public perception, security) while maximizing performance measures, i.e. transparency of methodology, efficiency of operations and quality of data.

The analysis also checked the machine learning models, working environment, algorithms used in order to provide recommendations. Particular attention was paid to the reproducibility of the results.

With the exception of a redundant over-fitting activity, the environment, the approach and the algorithms used seem to be adequate to meet the needs of the project and its launch into production. In future iterations, a more balanced focus on model variability and bias would be beneficial. The overall approach used to ensure reproducibility of results is excellent, simple and clear.

7.1. Environment

The SPSS environment is used for all processing, modeling and deployment. This environment is stable and well supported. Its disadvantage is its lack of flexibility; the options are limited to what SPSS offers.

7.2. Framework

The challenge is to classify visa applications for China. There are three years of labeled data (approved or not). Data training is done in batches and performance is measured using the accuracy of predicting approved visas. The approach used follows standards used in the field of machine learning.

7.3. Small Population (Imbalance)

The number of visas refused is relatively small compared to that of visas approved. The undersampling techniques used to reduce this issue are excellent. Given its reducing impact on sample size, it would be useful to better understand the effect of this imbalance. Only one approach has been tested.

7.4. Modeling

The modeling uses three years of data. The training part is made up of applications from the three years. The cross-validation part (called testing) is composed of applications from the last year only. And the testing part (called validation) is made up of months that are neither training nor cross-validation.

The modeling technique used for this project is a decision tree which is an excellent tool recognized for its transparency but relatively less precise than others. On the other hand, some modeling techniques could be improved and others canceled. Better representation of the cross-validation part to the training part could give better results. Using random numbers (seed) to determine the best model is unnecessary and puts unnecessary pressure on over-fitting. It seems that older data has less value than newer data. These two issues show that actions linked to modeling are likely to reduce the bias in favor of the variance of the model.

7.5. Retraining

The decision to retrain the models every three months is excellent.

7.6. Reproducibility

The use of SPSS software, the documentation and justification of each variable, the use of "seed", the transparency of the chosen model are excellent for reproducibility of results and transfer of knowledge. The risk of issues related to reproducibility is reduced to a minimum.

8. Recommendations

Following the two days of consultations, nine recommendations were identified.

1. Have a strategy for monitoring the retraining and updating parameters as needed.

   IRCC Response: Key performance metrics are monitored weekly by multiple stakeholders. Retraining is performed if a deterioration in these metrics indicates a need to update the model.

2. Change the unit of analysis to "family" instead of "person".

   IRCC Response: IRCC has maintained an analysis based on individual clients because this approach is more comprehensive. When multiple clients travel together as a group, the analysis takes into account information about each member of the group to inform the triage.

3. Standardize the training to the "Validation" group (also called Testing in the literature) so that the training data is similar to the test data.

   IRCC Response: IRCC modified its approach following the NRC review to support greater representativity of the validation data. This sample now encompasses applications from various parts of the year to ensure that seasonal patterns in intake are captured. The validation data is also selected among the most recent applications processed to ensure that the model accurately generalizes to the most recent trends in the program.

4. In a parallel environment, develop more sophisticated models starting with: Random Forest, support vector machine (SVM), Neural network, Deep learning, etc.

   IRCC Response: Some experiments were undertaken with alternative model types in our testing environment (never deployed). Gains in accuracy were marginal and came at the expense of a significant loss in the ability to explain the model's predictions. This would not have been acceptable in light of IRCC's efforts to use technology responsibly. Consequently, decision trees were maintained as the model type for this project.

5. Increase, or even double or triple, the size of the training group (approved) even if the number of refusals is relatively small.

   IRCC Response: Some experiments were undertaken with a larger representation of approved applications in the training sample. However, the best models were obtained when the training data was roughly balanced between approved and refused applications.

6. Increase the sample by combining data from other countries of origin.

IRCC Response: This approach was extended to India in 2018 and to all other countries in 2022.

7. Use of "clustering" to better understand the differences between seasons, countries, etc. and fraud analysis.

   IRCC Response: Most of the data used to train the model is non-numerical, which makes them less conducive to clustering techniques. IRCC experimented with Gower distances that accommodate non-numerical data, but the results were not helpful. Other tools have been developed for fraud analysis.

8. Consider the use of free software such as R or Python to experiment with new approaches and have more flexibility in modeling.

   IRCC Response: IRCC is migrating all of its models to Python to reduce costs, increase functionality and improve the productivity of analytics team.

9. Experiment with different forms of imbalance. Identify the approach that optimizes the accuracy of the results.

   IRCC Response: IRCC has experimented with different approaches to address imbalance in the training data. The best models were obtained when the training data was roughly balanced between approved and refused applications.

9. Future Collaboration

Different opportunities for collaboration between IRCC and the NRC were also discussed:

1. *(Redacted in accordance with Section 16 of the Access to Information Act).*
2. Use NRC calculation platforms to test more complex models.
3. Develop code for more sophisticated models (in R or Python for example).